

From Discrete Model to Continuum Limit

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* “paper” in the following content refers to [Anguige's paper](#)

Section 1. Deriving Continuum Limit from Discrete Model

Our original (discrete) conservation equation is the following: (equation 4 in the paper)

$$\frac{\partial \rho_i}{\partial t} = \frac{J_{i-\frac{1}{2}} - J_{i+\frac{1}{2}}}{h}$$

$J_{i+\frac{1}{2}}$ is the net flux of particles from x_i to x_{i+1} , where (equation 7 in the paper, from 4 to 7 will be verified in section 3 in this report)

$$J_{i+\frac{1}{2}} = \frac{R_i - R_{i+1}}{h} \quad \text{with } R_i = \rho_i(1 - \alpha + \alpha(\rho_{i-1}\rho_{i+1} - \rho_{i-1} - \rho_{i+1} + 1))$$

Our goal here is to prove the limit of this discrete model is the following continuum form:

$$\frac{\partial \rho}{\partial t} = \frac{\partial^2 K}{\partial x^2}$$

Where $K(\rho) = \rho(1 - 2\alpha\rho + \alpha\rho^2)$ is defined in the paper with diffusivity $D(\rho) = K'(\rho)$, and the “limit” means $h \rightarrow 0$, i.e. the width of spatial grid approaches to 0, as the time was not been discretized in our model.

Let's start with simplifying the expression of R_i for a little bit:

$$\begin{aligned} R_i &= \rho_i(1 - \alpha + \alpha(\rho_{i-1}\rho_{i+1} - \rho_{i-1} - \rho_{i+1} + 1)) \\ &= \rho_i(1 - \alpha + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1} - \alpha\rho_{i+1} + \alpha) \\ &= \rho_i(1 + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1} - \alpha\rho_{i+1}) \end{aligned}$$

Then, we convert the above equality into the expression of R_i only w.r.t ρ_i by Taylor expansions:

$$\begin{aligned} \rho_{i-1} &= \rho(x_{i-1}) = \rho(x_i - h) = \rho(x_i) - h\rho'(x_i) + \frac{h^2}{2}\rho''(x_i) - \frac{h^3}{6}\rho'''(x_i) + O(h^4) \\ &= \rho_i - h\rho'_i + \frac{h^2}{2}\rho''_i - \frac{h^3}{6}\rho'''_i + O(h^4) \end{aligned}$$

Notes that ρ' represents $\partial_x \rho_i$, the partial derivative w.r.t space variable x , and similarly for ρ'_i , ρ''_i , etc. The same, for ρ_{i+1} , we have

$$\rho_{i+1} = \rho_i + h\rho'_i + \frac{h^2}{2}\rho''_i + \frac{h^3}{6}\rho'''_i + O(h^4)$$

Plug in the discrete expression of R_i , we see

$$\begin{aligned} R_i &= \rho_i \left(1 - 2\alpha\rho_i + \alpha\rho_i^2 + \alpha h^2(\rho_i\rho''_i - \rho_i'^2 - \rho_i'') \right) + O(h^4) \\ &= \rho_i(1 - 2\alpha\rho_i + \alpha\rho_i^2) + \alpha h^2(\rho_i^2\rho''_i - \rho_i\rho_i'^2 - \rho_i\rho_i'') + O(h^4) \\ &= K(\rho_i) + h^2 H(\rho_i) + O(h^4) \end{aligned}$$

Here recall that $K(\rho) = \rho(1 - 2\alpha\rho + \alpha\rho^2)$ as we mentioned before, and

$$H(f) := \alpha(f^2 f'' - f f'^2 - f f'')$$

Again, f' represents $\frac{\partial f}{\partial x}$, etc.

In fact, $R_i = K(\rho_i) + h^2 H(\rho_i) + h^4(\dots) + h^6(\dots) + \dots$, by cancellation between two Taylor expansions, only h^k terms with even k left. To achieve better performance w.r.t accuracy i.e. modified equation, we need to specify the expression of $O(h^4)$ term here, and stop by $O(h^6)$ to represents higher order terms.

Similarly, we can get

$$\begin{aligned} R_{i-1} &= K(\rho_{i-1}) + h^2 H(\rho_{i-1}) + O(h^4) \\ R_{i+1} &= K(\rho_{i+1}) + h^2 H(\rho_{i+1}) + O(h^4) \end{aligned}$$

Now let's turn back to our discrete conservation equation:

$$\begin{aligned} \frac{\partial \rho_i}{\partial t} &= \frac{J_{i-\frac{1}{2}} - J_{i+\frac{1}{2}}}{h} \\ &= \frac{\frac{R_{i-1} - R_i}{h} - \frac{R_i - R_{i+1}}{h}}{h} \\ &= \frac{1}{h^2} (R_{i-1} + R_{i+1} - 2R_i) \\ &= \frac{1}{h^2} (K(\rho_{i-1}) + K(\rho_{i+1}) - 2K(\rho_i)) + (H(\rho_{i-1}) + H(\rho_{i+1}) - 2H(\rho_i)) + O(h^2) \\ &= \frac{\frac{K(\rho_{i-1}) - K(\rho_i)}{h} - \frac{K(\rho_i) - K(\rho_{i+1})}{h}}{h} + (H(\rho_{i-1}) + H(\rho_{i+1}) - 2H(\rho_i)) + O(h^2) \\ &= \quad \quad \quad (part\ 1) \quad \quad \quad + \quad \quad \quad (part\ 2) \quad \quad \quad + (part\ 3) \end{aligned}$$

For part 1 above, simply by finite difference,

$$\lim_{h \rightarrow 0} \frac{\frac{K(\rho_{i-1}) - K(\rho_i)}{h} - \frac{K(\rho_i) - K(\rho_{i+1})}{h}}{h} = \frac{\partial^2 K}{\partial x^2}$$

Or more explicitly, to show the accuracy of the scheme, we say

$$\frac{\frac{K(\rho_{i-1}) - K(\rho_i)}{h} - \frac{K(\rho_i) - K(\rho_{i+1})}{h}}{h} = \frac{\partial^2 K}{\partial x^2} + O(h^2)$$

For part 2:

$$H(\rho_{i-1}) = \alpha(\rho_{i-1}^2 \rho_{i-1}'' - \rho_{i-1} \rho_{i-1}'^2 - \rho_{i-1} \rho_{i-1}'')$$

Again, we hope to get the consistent expressions on both $H(\rho_{i-1})$ and $H(\rho_{i+1})$ w.r.t only ρ_i , in order to see further cancellation. So by Taylor expansion, $\rho_{i-1} = \rho_i + O(h)$, $\rho_{i-1}' = \rho_i' + O(h)$ and $\rho_{i-1}'' = \rho_i'' + O(h)$, we plug them into $H(\rho_{i-1})$:

$$H(\rho_{i-1}) = \alpha(\rho_i^2 \rho_i'' - \rho_i \rho_i'^2 - \rho_i \rho_i'') + O(h^2) = H(\rho_i) + O(h^2)$$

Similarly,

$$H(\rho_{i+1}) = \alpha(\rho_i^2 \rho_i'' - \rho_i \rho_i'^2 - \rho_i \rho_i'') + O(h^2) = H(\rho_i) + O(h^2)$$

So the part 2 becomes

$$H(\rho_{i-1}) + H(\rho_{i+1}) - 2H(\rho_i) = O(h^2)$$

Overall, the combination of those 3 parts:

$$\frac{\partial \rho_i}{\partial t} = \frac{\partial^2 K}{\partial x^2} + O(h^2) \rightarrow \frac{\partial^2 K}{\partial x^2} \quad \text{as } h \rightarrow 0$$

That is, we get exactly the following desired continuum limit equation:

$$\frac{\partial \rho_i}{\partial t} = \frac{\partial^2 K}{\partial x^2}$$

We can see from the analysis that this finite difference scheme is of 2nd order accuracy. To achieve higher order accuracy, let's consider the modified equation in next section.

Section 2. Deriving Modified Equation

Recall $R_i = K(\rho_i) + h^2 H(\rho_i) + O(h^4)$, where those h comes from the Taylor expansion of ρ_{i-1} and ρ_{i+1} . In our preceding analysis, we let those h tends to 0 at the same time with the h from our original discrete model, that is, by taking the limit simultaneously. Now, instead of taking the limit, we use a finite difference error “tail” e_h to write equality between discrete equation and its continuum limit, i.e.

$$\frac{\partial \rho_i}{\partial t} = \text{discrete equation} = \text{continuum limit} + e_h$$

So

$$\begin{aligned} \frac{\partial \rho_i}{\partial t} &= \frac{J_{i-\frac{1}{2}} - J_{i+\frac{1}{2}}}{h} \\ &= \frac{\frac{R_{i-1} - R_i}{h} - \frac{R_i - R_{i+1}}{h}}{h} \\ &= \frac{R_{i-1} + R_{i+1} - 2R_i}{h^2} \\ &= \frac{(K(\rho_{i-1}) + K(\rho_{i+1}) - 2K(\rho_i)) + h^2(H(\rho_{i-1}) + H(\rho_{i+1}) - 2H(\rho_i)) + O(h^4)}{h^2} \\ &= \frac{\partial^2(K(\rho) + h^2 H(\rho) + O(h^4))}{\partial x^2} + e_h \end{aligned}$$

Now, let's specify the tail term e_h , which is the error comes from the finite difference scheme applied on $\frac{\partial^2 K}{\partial x^2}$:

$$\frac{\partial^2 K}{\partial x^2} = \frac{\frac{K(\rho_{i-1}) - K(\rho_i)}{h} - \frac{K(\rho_i) - K(\rho_{i+1})}{h}}{h} + e_h$$

Here, we can still use Taylor expansion to analyze the error:

$$\begin{aligned} K(\rho_{i-1}) &= K(\rho_i) - h \frac{\partial K(\rho_i)}{\partial x} + \frac{h^2}{2} \frac{\partial^2 K(\rho_i)}{\partial x^2} - \frac{h^3}{6} \frac{\partial^3 K(\rho_i)}{\partial x^3} + \frac{h^4}{24} \frac{\partial^4 K(\rho_i)}{\partial x^4} - \frac{h^5}{120} \frac{\partial^5 K(\rho_i)}{\partial x^5} \\ &\quad + O(h^6) \\ K(\rho_{i+1}) &= K(\rho_i) + h \frac{\partial K(\rho_i)}{\partial x} + \frac{h^2}{2} \frac{\partial^2 K(\rho_i)}{\partial x^2} + \frac{h^3}{6} \frac{\partial^3 K(\rho_i)}{\partial x^3} + \frac{h^4}{24} \frac{\partial^4 K(\rho_i)}{\partial x^4} + \frac{h^5}{120} \frac{\partial^5 K(\rho_i)}{\partial x^5} \\ &\quad + O(h^6) \end{aligned}$$

$$\begin{aligned}
\frac{\frac{K(\rho_{i-1}) - K(\rho_i)}{h} - \frac{K(\rho_i) - K(\rho_{i+1})}{h}}{h} &= \frac{1}{h^2} (K(\rho_{i-1}) + K(\rho_{i+1}) - 2K(\rho_i)) \\
&= \frac{1}{h^2} \left(h^2 \frac{\partial^2 K(\rho_i)}{\partial x^2} + \frac{h^4}{12} \frac{\partial^4 K(\rho_i)}{\partial x^4} + O(h^6) \right) \\
&= \frac{\partial^2 K(\rho_i)}{\partial x^2} + \frac{h^2}{12} \frac{\partial^4 K(\rho_i)}{\partial x^4} + O(h^4)
\end{aligned}$$

Hence,

$$e_h = \frac{h^2}{12} \frac{\partial^4 K(\rho)}{\partial x^4} + O(h^4)$$

Then, we can back to

$$\begin{aligned}
\frac{\partial \rho_i}{\partial t} &= \frac{J_{i-\frac{1}{2}} - J_{i+\frac{1}{2}}}{h} \\
&= \frac{\partial^2 (K(\rho) + h^2 H(\rho) + O(h^4))}{\partial x^2} + e_h \\
&= \frac{\partial^2 (K(\rho) + h^2 H(\rho) + O(h^4))}{\partial x^2} + \frac{h^2}{12} \frac{\partial^4 K(\rho)}{\partial x^4} + O(h^4) \\
&= \frac{\partial^2 \left(K(\rho) + h^2 H(\rho) + \frac{h^2}{12} \frac{\partial^4 K(\rho)}{\partial x^4} \right)}{\partial x^2} + O(h^4)
\end{aligned}$$

Which is exactly the modified equation.

To see this clearer, let's rewrite $H(\rho)$ in formal way:

$$\begin{aligned}
H(\rho) &= \alpha(\rho^2 \rho'' - \rho \rho'^2 - \rho \rho'') \\
&= \alpha \rho^2 \frac{\partial^2 \rho}{\partial x^2} - \alpha \rho \left(\frac{\partial \rho}{\partial x} \right)^2 - \alpha \rho \frac{\partial^2 \rho}{\partial x^2} \\
&= \alpha \rho \frac{\partial^2 \rho}{\partial x^2} (\rho - 1) - \alpha \rho \left(\frac{\partial \rho}{\partial x} \right)^2
\end{aligned}$$

then

$$\begin{aligned}
\frac{\partial \rho_i}{\partial t} &= \frac{\partial^2}{\partial x^2} \left(K(\rho) + h^2 \left(H(\rho) + \frac{1}{12} \frac{\partial^4 K(\rho)}{\partial x^4} \right) \right) + O(h^4) \\
&= \frac{\partial^2}{\partial x^2} \left(K(\rho) + h^2 \left(\alpha \rho \frac{\partial^2 \rho}{\partial x^2} (\rho - 1) - \alpha \rho \left(\frac{\partial \rho}{\partial x} \right)^2 + \frac{1}{12} \frac{\partial^4 K(\rho)}{\partial x^4} \right) \right) + O(h^4)
\end{aligned}$$

Section 3. From (4) to (7): Verification of the Equivalence of 2 Discrete Expressions

Originally, we have:

$$J_{i+\frac{1}{2}} = \frac{1}{h} [\rho_i(1 - \rho_{i+1})(1 - \alpha\rho_{i-1}) - \rho_{i+1}(1 - \rho_i)(1 - \alpha\rho_{i+2})]$$

So

$$\begin{aligned} h \cdot J_{i+\frac{1}{2}} &= \rho_i(1 + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1} - \rho_{i+1}) - \rho_{i+1}(1 + \alpha\rho_i\rho_{i+2} - \alpha\rho_{i+2} - \rho_i) \\ &= \rho_i(1 + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1}) - \rho_i\rho_{i+1} - \rho_{i+1}(1 + \alpha\rho_i\rho_{i+2} - \alpha\rho_{i+2}) + \rho_{i+1}\rho_i \\ &= \rho_i(1 + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1}) - \rho_{i+1}(1 + \alpha\rho_i\rho_{i+2} - \alpha\rho_{i+2}) \\ &= \rho_i(1 + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1}) - \alpha\rho_i\rho_{i+1} - \rho_{i+1}(1 + \alpha\rho_i\rho_{i+2} - \alpha\rho_{i+2}) + \alpha\rho_{i+1}\rho_i \\ &= \rho_i(1 + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1} - \alpha\rho_{i+1}) - \rho_{i+1}(1 + \alpha\rho_i\rho_{i+2} - \alpha\rho_{i+2} - \alpha\rho_i) \\ &= R_i - R_{i+1} \end{aligned}$$

hence

$$J_{i+\frac{1}{2}} = \frac{R_i - R_{i+1}}{h} \quad \text{with } R_i = \rho_i(1 + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1} - \alpha\rho_{i+1})$$

Section 4. Try Higher Order Taylor Expansion at ρ_i for R_i

Let's back to

$$R_i = \rho_i(1 + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1} - \alpha\rho_{i+1})$$

Now, take Taylor expansion again, but truncate at higher order term:

$$\begin{aligned} \rho_{i-1} &= \rho_i - h\rho'_i + \frac{h^2}{2}\rho''_i - \frac{h^3}{6}\rho'''_i + \frac{h^4}{24}\rho^{(4)}_i - \frac{h^5}{120}\rho^{(5)}_i + O(h^6) \\ \rho_{i+1} &= \rho_i + h\rho'_i + \frac{h^2}{2}\rho''_i + \frac{h^3}{6}\rho'''_i + \frac{h^4}{24}\rho^{(4)}_i + \frac{h^5}{120}\rho^{(5)}_i + O(h^6) \end{aligned}$$

where $\rho_i^{(4)}$ represents $\frac{\partial^4 \rho_i}{\partial x^4} = \frac{\partial^4 \rho(x_i)}{\partial x^4}$, etc. Take 2 series above into R_i , we have:

$$\begin{aligned} R_i &= \rho_i - 2\alpha\rho_i^2 + \alpha\rho_i^3 + h^2(\alpha\rho_i^2\rho''_i - \alpha\rho_i\rho_i'^2 - \alpha\rho_i\rho''_i) \\ &\quad + h^4\left(\frac{1}{4}\alpha\rho_i\rho_i''^2 - \frac{1}{3}\alpha\rho_i\rho_i'\rho_i''' - \frac{1}{12}\alpha\rho_i\rho_i^{(4)} + \frac{1}{12}\alpha\rho_i^2\rho_i^{(4)}\right) + O(h^6) \end{aligned}$$